Ahmedabad University

Project 2 - Person Retrieval

End-semester Project Presentation

Group - 15

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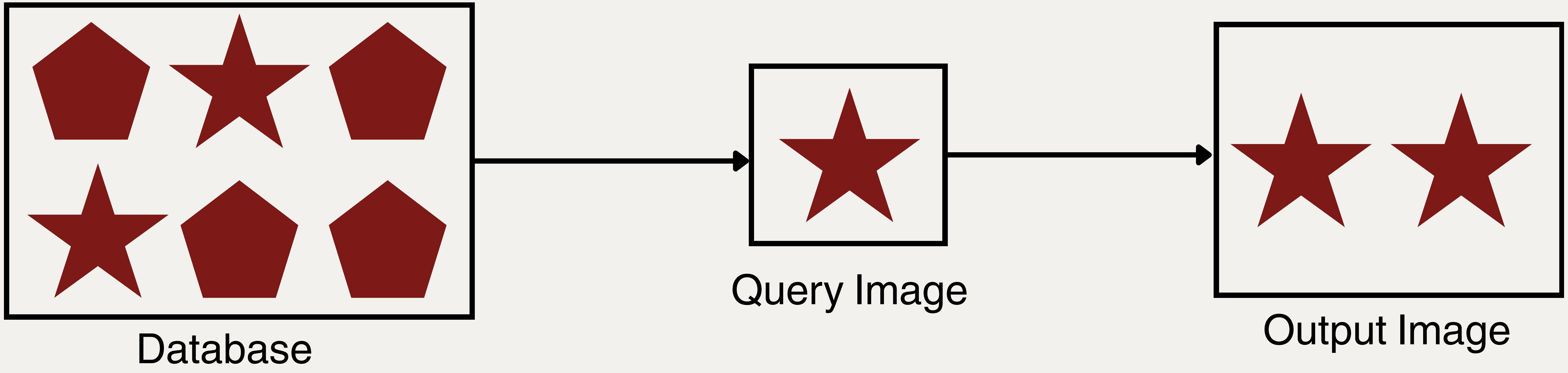
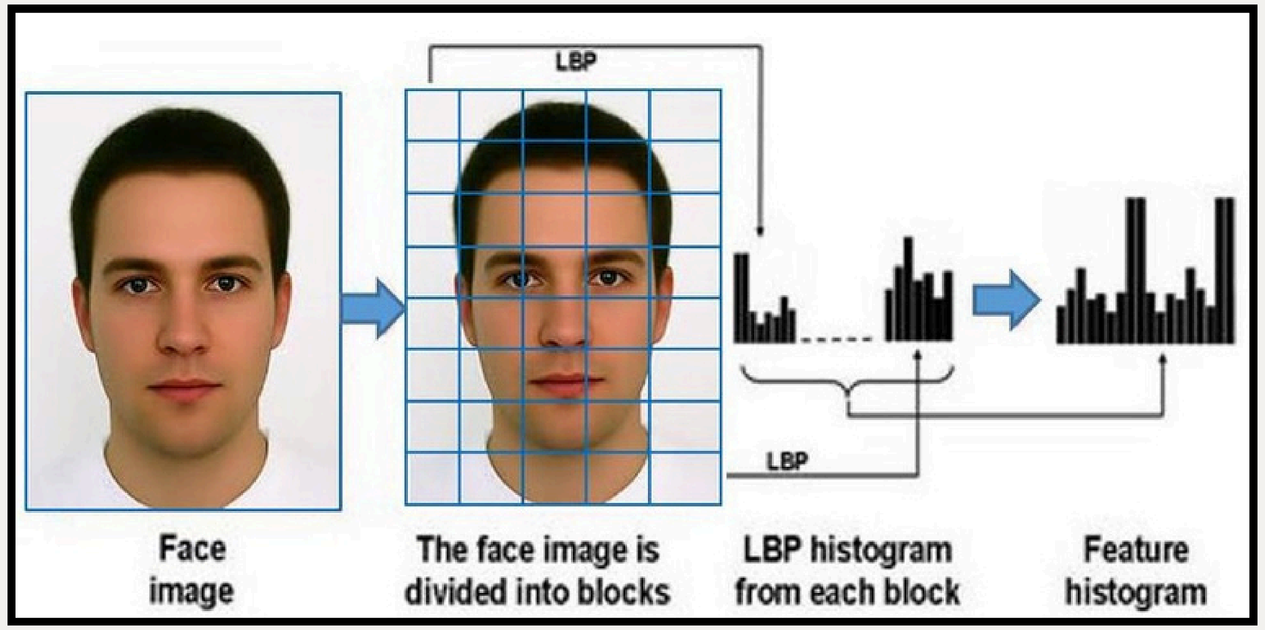
Introduction

- Person Retrieval: finding all images of the same person in the given photo
- Normally done with **Machine Learning / Deep Learning**
- Our approach: **Only Classical Digital Image Processing (DIP) Techniques**
- Our Database: **20 photos** → **4 persons x 5 photos each**

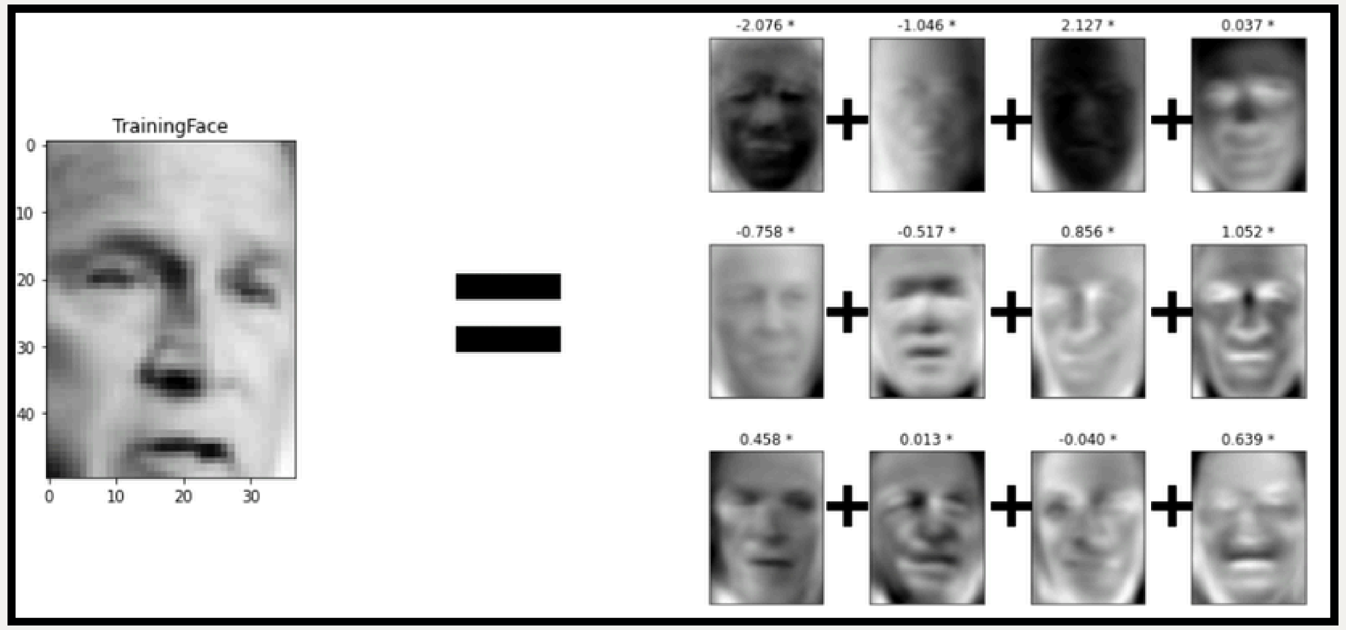


Old Methodology

- **Preprocessing:**
  - Grayscale, Resize, Histogram Equalization
- **Feature Extraction:**
  - Local Binary Pattern (LBP) - texture
  - Discrete Cosine Transform (DCT) - frequency
  - Principal Component Analysis (PCA) (Eigenfaces) - dimensionality reduction



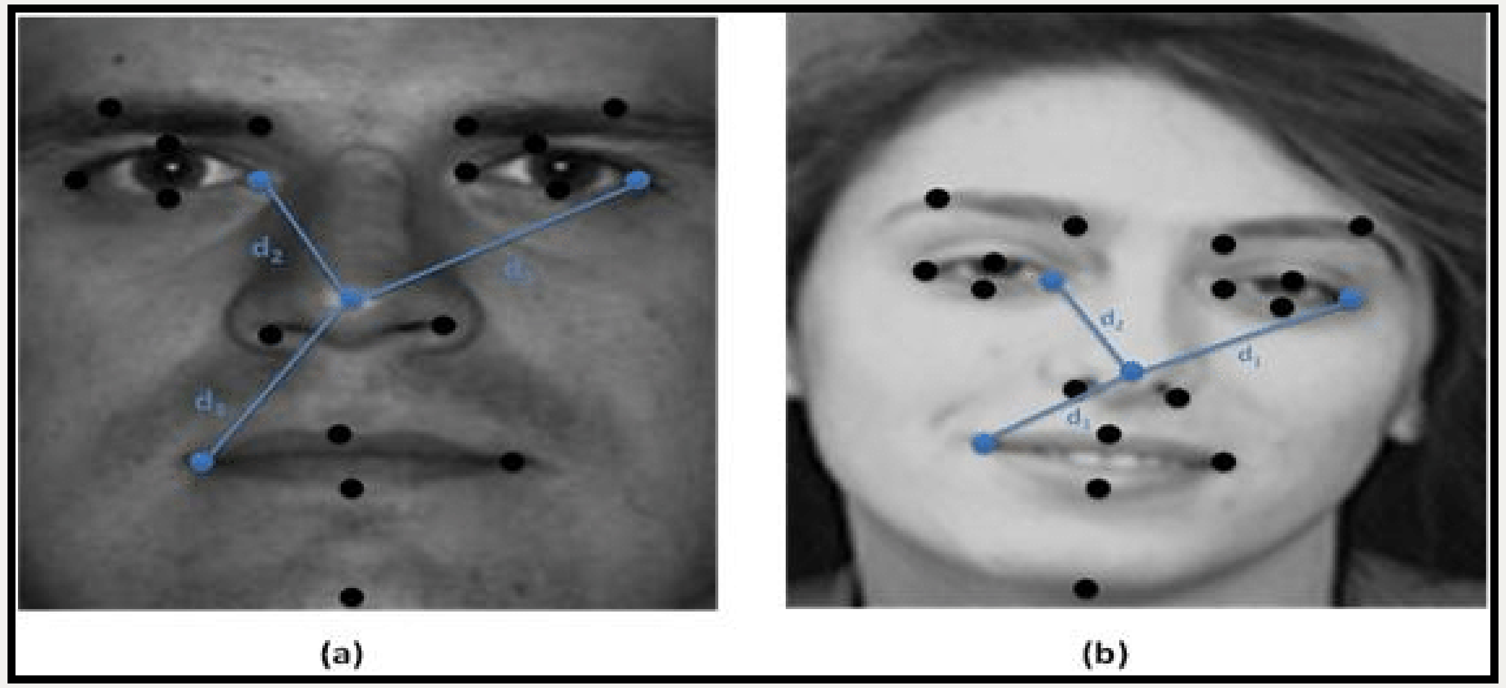
Local Binary Pattern



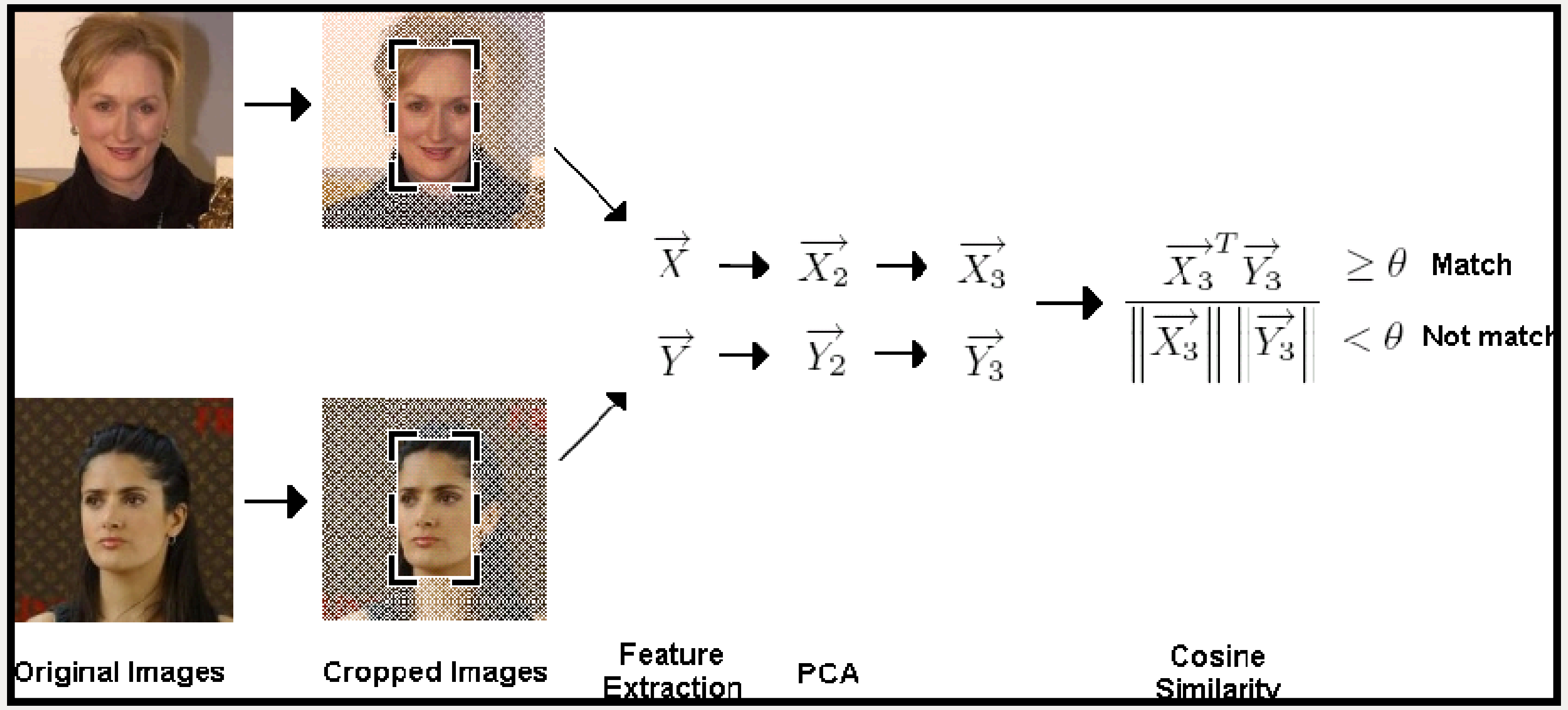
Principal Component Analysis

Old Methodology

- **Similarity Matching:**
  - Euclidean Distance
  - Cosine Similarity



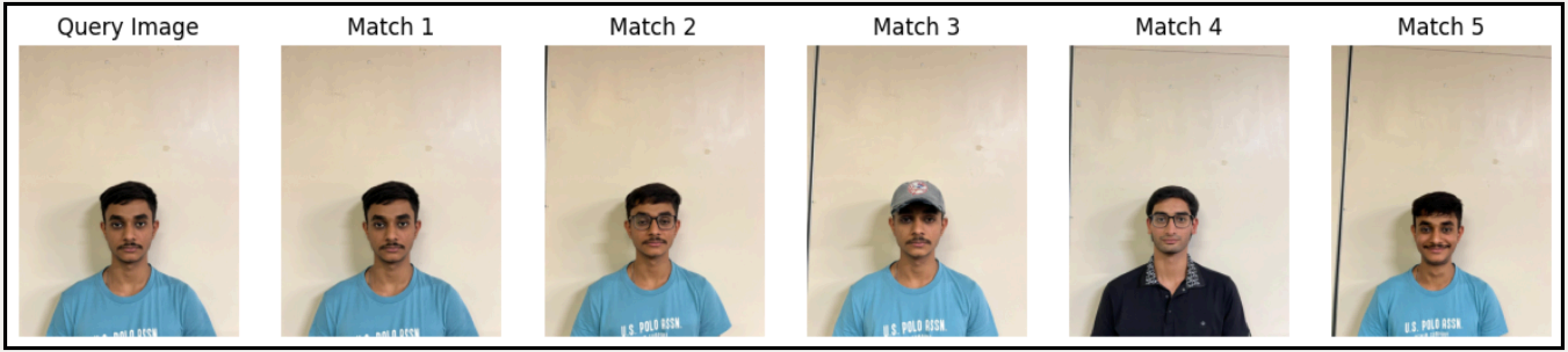
Euclidean Distance



Cosine Similarity

Conclusion (Work before Mid-sem)

- The project successfully demonstrates Face Retrieval using only classical Digital Image Processing techniques.
- By combining color histograms, edge maps, and texture features (LBP), the system achieves reliable identification of visually similar faces.
- Overall, the code was working but was not accurate.



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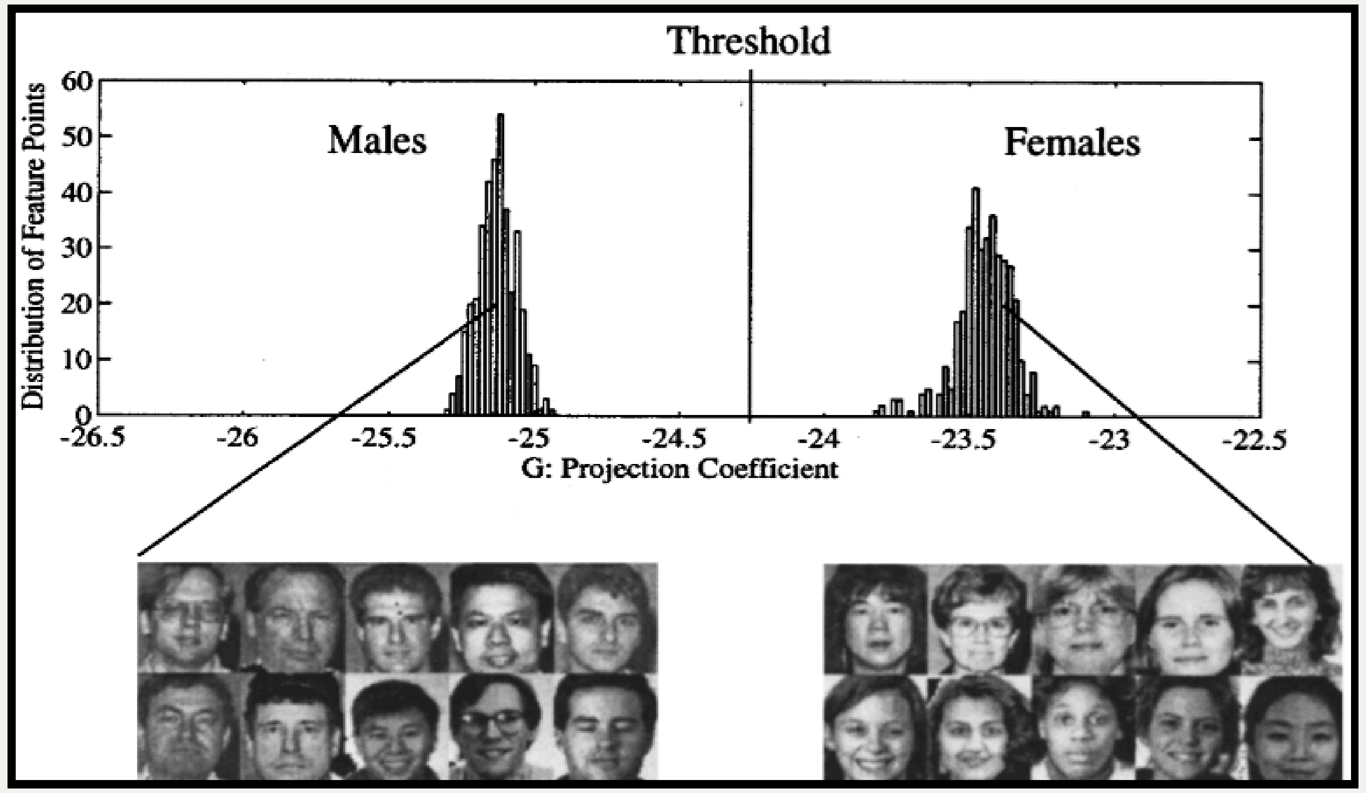
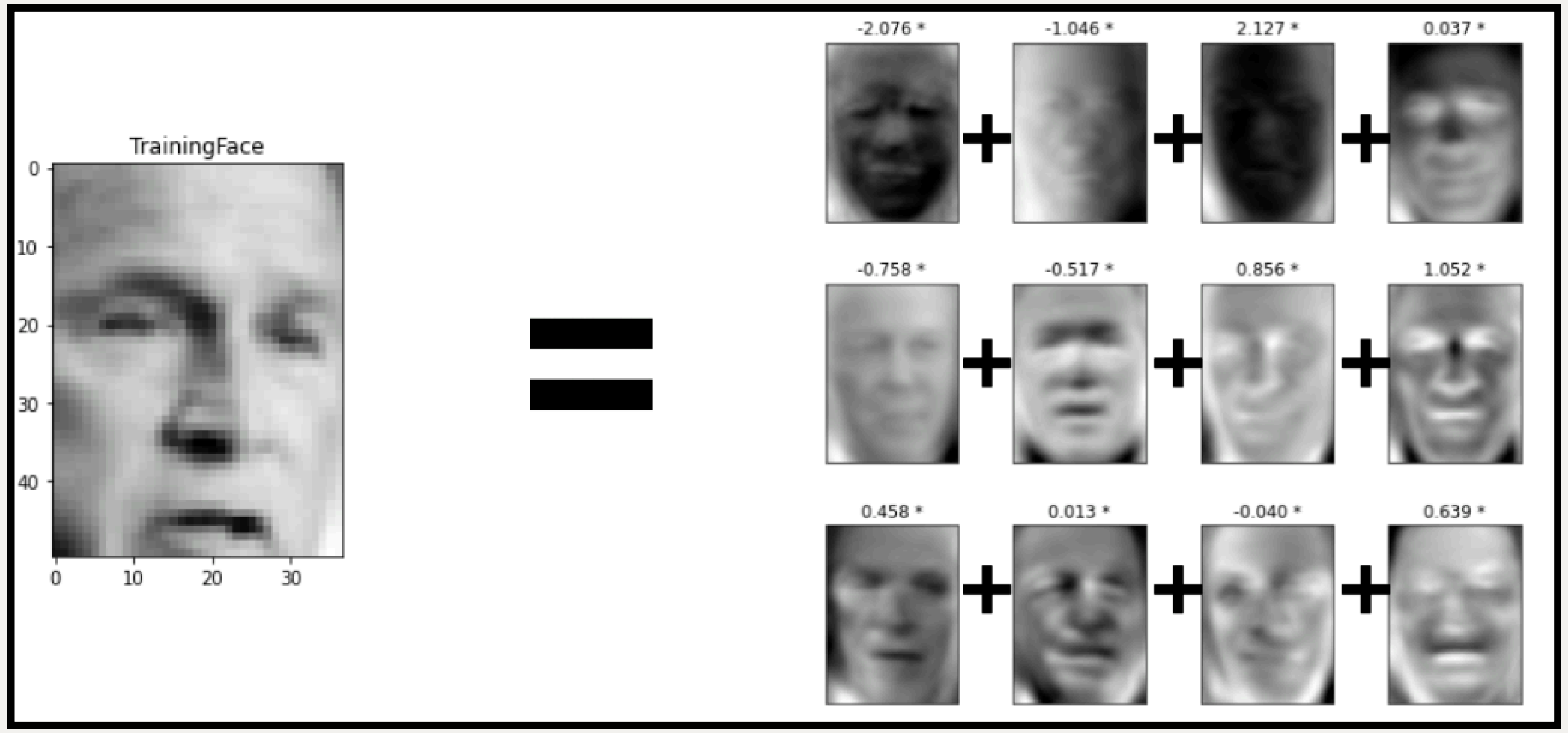
Updated Methodology

- **Preprocessing:**
  - Grayscale, Data Vectorization and Label Creation
- **Feature Extraction:**
  - Linear Discriminant Analysis (LDA)
  - Principal Component Analysis (PCA) - dimensionality reduction



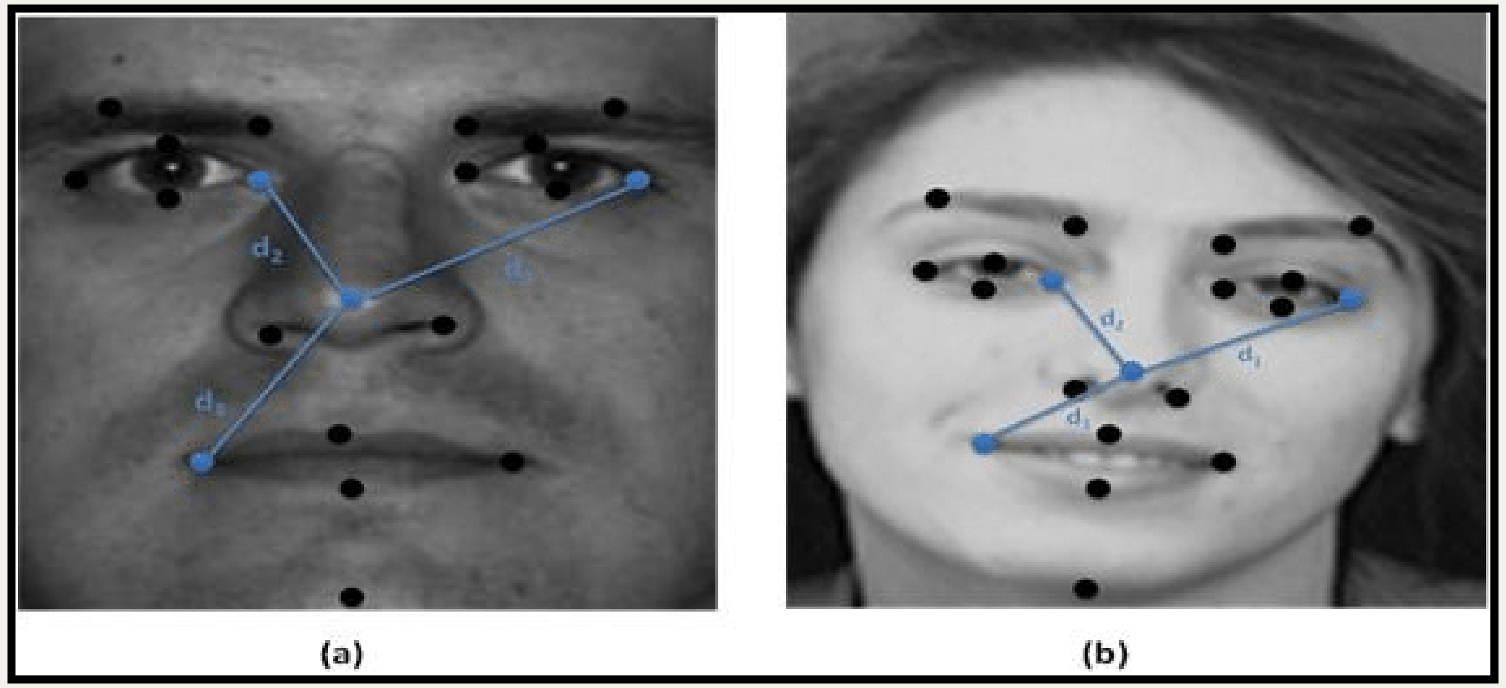
Linear Discriminant Analysis



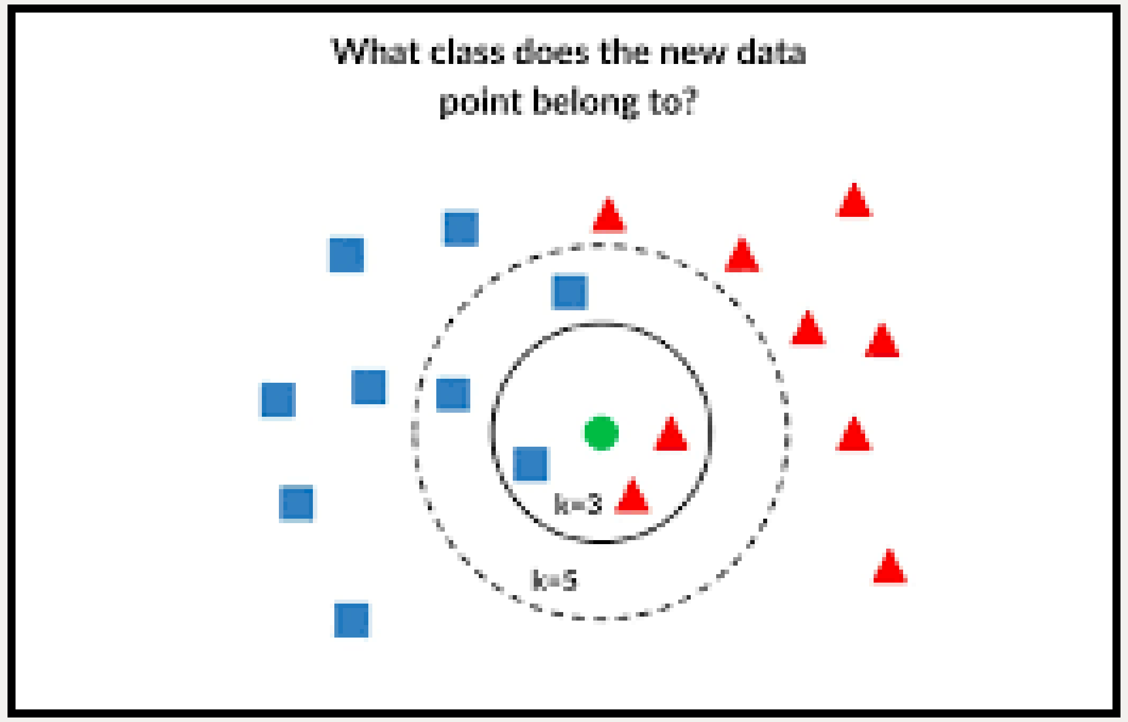
Principal Component Analysis

Updated Methodology

- **Similarity Matching:**
  - Euclidean Distance
  - K-Nearest Neighbor (KNN) Retrieval



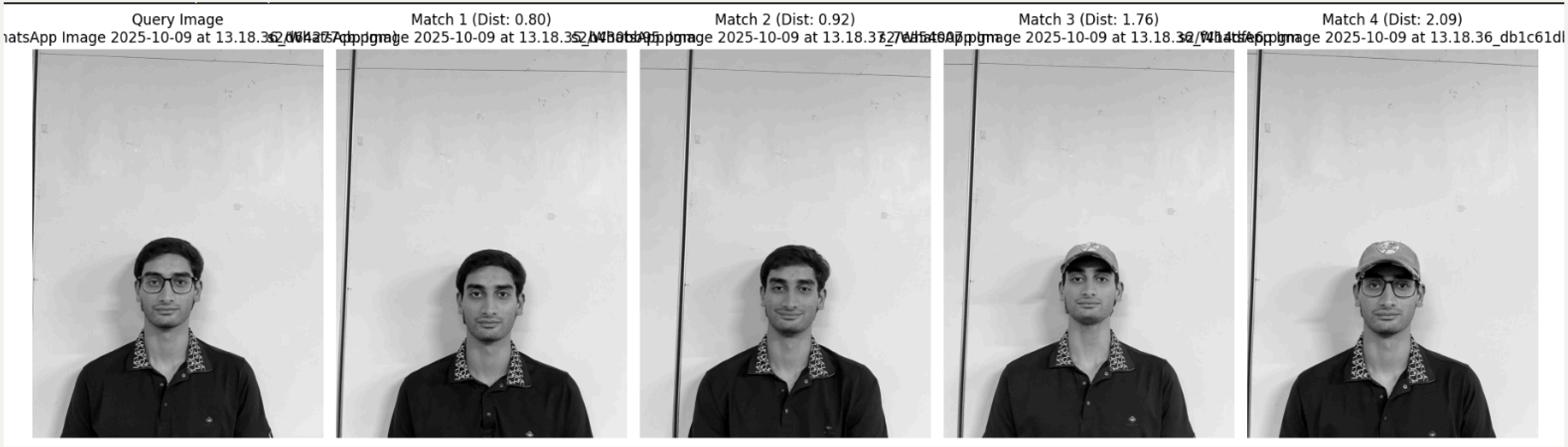
Euclidean Distance



KNN Retrieval

Results

- Query image and retrieved results displayed side-by-side
- Output shows **4 matching images**
- Retrieval accuracy with change in methodology
- Performance depends on lighting variations



Conclusion

- The project successfully demonstrates a complete Person Retrieval system using only classical Digital Image Processing techniques.
- By upgrading the methodology from mid-semester work, the integration of PCA + LDA significantly improved feature discrimination and retrieval accuracy.
- The system now provides more reliable matching using Euclidean distance and KNN-based retrieval, even with slight variations in lighting and pose.
- Experimental results show that the updated pipeline is more efficient, consistent, and scalable for small datasets.

References

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